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Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #26 | Smart Cities, Transport & Logistics

Peer-to-Peer Carpooling & Expense Splitter

<i>Req ID</i>	<i>Type</i>	<i>Description</i>	<i>Priority</i>	<i>Acceptance Criteria</i>	<i>Rationale</i>
FR-001	Functional	The system shall allow a Driver Host to publish a trip by providing source, destination, travel time, and available seats.	High	Pass: A ride with all mandatory details is successfully published and made available for matching. Fail: A ride is published when one or more mandatory details are missing.	Ride details are required before the system can match drivers with riders.
FR-002	Functional	The system shall allow a Rider Commuter to search for available trips using source, destination, and preferred travel time.	High	Pass: The system displays available rides that satisfy the entered trip details and route-compatibility rules. Fail: An incompatible ride is shown as a valid match.	Enables riders to find suitable carpool options for their commute.
FR-003	Functional	The system shall calculate route compatibility between driver and passenger commute paths, permitting matches only within a 1.5 km detour tolerance.	High	Pass: Route match confirmed and fuel cost split calculated Fail: Detour exceeding tolerance accepted	Ensures that drivers are not required to make excessive deviations from their regular route.
FR-004	Functional	The system shall automatically calculate and display the Rider Commuter's fuel-cost share for a confirmed carpool trip.	High	Pass: For predefined trip and fuel-cost values, the calculated share matches the expected result according to the defined cost-sharing rule. Fail: The calculated amount differs from the expected value.	Automated fuel-cost splitting is one of the core functions of the system.
FR-005	Functional	The system shall allow a Rider Commuter to view the Driver Host's rating and driving-license verification status before requesting a ride.	Medium	Pass: The rider can view the available driver rating and verification status before sending a ride request. Fail: The system marks an unverified driver as verified or hides the available verification status from the rider.	Helps riders make more informed and safer decisions when selecting a carpool.

NFR-001	Nonfunctional	The system shall securely store user ratings, driving-license verification documents, and travel histories using encryption and encrypted backups.	High	Pass: Security testing verifies that protected user information and backups are stored in encrypted form. Fail: Any protected information or backup is stored unencrypted.	Protects sensitive identity, reputation, and travel information from unauthorized access.
NFR-002	Nonfunctional	The system shall return compatible ride-search results within 3 seconds under the defined performance-test workload.	Medium	Pass: Search results are returned within 3 seconds during the defined performance test. Fail: The response time exceeds 3 seconds.	Ensures that commuters can search for rides without unnecessary delay.